

Homework 22

Please read Section 6.4 through the end of the discussion of z-scores following Example 6.24.

Please work the following exercises from Chapter 6 of Durrett. Show your steps in the solutions, which will involve calculating expected value (μ), standard deviation (σ), and z-scores and then looking up suitable standard normal probabilities. All the random variables in these questions are discrete, and so you should use the histogram correction.

21 (Textbook answer: 0.484)

Let N be the random variable representing the number of heads flipped in 100 tosses. Then N has the binomial(100, 0.5) distribution, and so $E[N] = 50$ and $\text{Var}(N) = 100(0.5)(0.5) = 25$. Since N is a sum of 100 iid indicator variables (indicating whether flip i is a head), we treat N as approximately normally distributed (with the normal(50, 25^2) distribution).

We are interested in the event $A = (N \geq 54) \cup (N \leq 46)$. We have $P(A) = P(N \geq 54) + P(N \leq 46)$ since the two events are disjoint. To apply the CLT approximation, we use the histogram correction and estimate $P(N \geq 53.5)$. The z-score for 53.5 is $(53.5 - 50)/5 = 3.5/5 = 0.70$. We thus have $P(N \geq 53.5) \approx P(\text{standard normal} \geq 0.7) = 1 - P(\text{standard normal} \leq 0.7)$ (since the normal distribution is continuous). The standard normal table gives $P(\text{standard normal} \leq 0.7) \approx 0.7580$, and so $P(N \geq 54) = P(N \geq 53.5) \approx 1 - 0.7580 = 0.2420$. Because the distribution of N is symmetric around 50, we have $P(N \leq 46) \approx 0.2420$, and so $P(A) \approx 0.4840$.

22 (Use the CLT!)

Let N again be the random variable representing the number of heads flipped in 100 tosses. We want to compare $P(N = 50)$ and $P(N \geq 60)$.

As above, we treat N as approximately normally distributed. We apply the histogram correction and view $P(N = 50)$ as $P(49.5 \leq N \leq 50.5)$. The z-score for 50.5 is $(50.5 - 50)/5 = 0.1$, and the z-score for 49.5 is $(49.5 - 50)/5 = -0.1$. The normal approximation for $P(N = 50)$ is this $P(-0.1 \leq \text{standard normal} \leq 0.1)$.

We calculate $P(-0.1 \leq \text{standard normal} \leq 0.1)$ as $P(\text{standard normal} \leq 0.1) - P(\text{standard normal} < -0.1)$. By the symmetry of the standard normal distribution around 0, $P(\text{standard normal} < -0.1) = P(\text{standard normal} > 0.1) = 1 - P(\text{standard normal} \leq 0.1)$. Therefore, $P(-0.1 \leq \text{standard normal} \leq 0.1) = 2P(\text{standard normal} \leq 0.1) - 1$. Using the normal table, we find $P(\text{standard normal} \leq 0.1) \approx 0.5398$, and so $P(-0.1 \leq \text{standard normal} \leq 0.1) \approx 0.0796$.

We approximate $P(N \geq 60)$ using the histogram correction and the CLT approximation (and continuity of the normal distribution):

$$P(N \geq 60) = P(N \geq 59.5) \approx P(\text{standard normal} \geq 1.9) = 1 - P(\text{standard normal} \leq 1.9) \approx 1 - 0.9713 = 0.0287.$$

It is much more likely to flip exactly 50 heads.

24

We treat the number N of games won in the season as a random variable with binomial(162, 0.5) distribution. Then $E[N] = 81$, $\text{Var}(N) = 162(0.5)(0.5) = 40.5$, and $\sigma(N) \approx 6.3640$. As above, we apply the

histogram correction and treat N as approximately normal. (The justification for the normality assumption is that N is a sum of 162 iid random variables, and so we expect it to have roughly a normal distribution by the CLT.)

$$P(N \geq 87) = P(N \geq 86.5) \approx P(\text{standard normal} \geq (86.5 - 81)/6.3640) = 1 - P(\text{standard normal} \leq 0.8642) \approx 1 - 0.8051 = 0.1949.$$

25 (Textbook answer: 0.0016)

We treat the number N of customers who pick United as a random variable with the binomial(400, 0.5) distribution. We thus have $E[N] = 200$, $\text{Var}(N) = 100$, and $\sigma(N) = 10$.

$$P(N \geq 230) = P(N \geq 229.5) \approx P(\text{standard normal} \geq 2.95) = 1 - P(\text{standard normal} \leq 2.95) \approx 1 - 0.9984 = 0.0016.$$

28

Let X be the random variable representing the number of 1s rolled in 600 tosses. We treat X as having the binomial(600, 1/6) distribution. We thus have $E[X] = 100$, $\text{Var}(X) = 600(1/6)(5/6) = 500/6$, and $\sigma(X) \approx 9.1287$. Since X is a sum of 600 iid indicator random variables (indicating whether roll i is a 1), by the CLT, we expect that X is approximately normally distributed.

$$P(90 \leq X \leq 110) = P(89.5 \leq X \leq 110.5) \approx P(-1.15 \leq \text{standard normal} \leq 1.15) = 2P(\text{standard normal} \leq 1.15) - 1 \approx 2(0.8749) - 1 = 0.7498.$$

30

We treat the number Y of red jelly beans in the bag as a random variable with the binomial(400, 0.1) distribution. We thus have $E[Y] = 400(0.1) = 40$, $\text{Var}(Y) = 400(0.1)(0.9) = 36$, and $\sigma(Y) = 6$. Since Y is a sum of 400 iid indicator variables (indicating “jelly bean i is red”), by the CLT, we expect that Y is approximately normally distributed.

$$P(Y \geq 45) = P(Y \geq 44.5) \approx P(\text{standard normal} \geq 0.75) = 1 - P(\text{standard normal} \leq 0.75) \approx 1 - 0.7734 = 0.2266.$$